REVISION WORKSHEET

UNITED ARAB EMIRATES
MINISTRY OF EDUCATION

الإمارات العربية المتحدة
وزارة التربية والتعليم

AL KHALIDIYA BOYS SCHOOL

Cycle 3 (10048) | Dibba Al Hisn | Sharjah Eastern

Practice Questions and Worked Answers

Grade 9 Advanced | Module 2 Lesson 2 | Where and When

All 30 questions are followed by answers. Study the reasoning, formula, substitution and units. Equivalent scientifically correct wording is acceptable.

Useful equations and guidance: Δx = xf − xi; Δt = tf − ti; distance adds path lengths; choose an origin and positive direction.

Term 1 2026–2027 | Textbook alignment: uploaded Grade 9 book, printed pages 32–36.

P01. Displacement is:

A. Total path length only B. Always positive
C. Change in position D. A scalar without direction

Answer

C. Change in position.

P02. Which is a vector quantity?

A. Mass B. Time
C. Distance D. Displacement

Answer

D. Displacement.

P03. A coordinate origin specifies:

A. The chosen zero position B. The direction of gravity everywhere
C. The object's speed D. The end of every trip

Answer

A. The chosen zero position.

P04. A negative position means the object is:

A. Necessarily moving left B. On the negative side of the chosen origin
C. Necessarily slowing down D. At a negative distance travelled

Answer

B. On the negative side of the chosen origin.

P05. An object returns to its starting point. Its net displacement is:

A. Equal to total distance B. Always positive
C. Zero D. Undefined

Answer

C. Zero.

P06. An object starts at x = -4 m and moves 10 m in the positive direction. Calculate its final position.

A. 6 m B. 12 m
C. 3 m D. -6 m

Answer

A. 6 m. $x_f = x_i + \Delta x = -4 + 10 = 6$ m.

P07. An object moves from x = 6 m to x = -4 m. Calculate its displacement.

A. -20 m B. -10 m
C. -5 m D. 10 m

Answer

B. -10 m. $\Delta x = x_f - x_i = -4 - 6 = -10$ m. The negative sign denotes the negative coordinate direction.

P08. A student walks 10 m east and then 4 m west. Taking east as positive, calculate the displacement.

A. 12 m B. 3 m
C. 6 m D. -6 m

Answer

C. 6 m. $\Delta x = +10 - 4 = +6$ m, east.

P09. An observation starts at t = 4 s and ends at t = 14 s. Calculate the time interval.

A. 20 s B. 5 s
C. -10 s D. 10 s

Answer

D. 10 s. $\Delta t = t_f - t_i = 14 - 4 = 10$ s.

P10. A student walks 10 m east and then 4 m west. Calculate the total distance travelled.

A. 14 m B. 28 m
C. 7 m D. -14 m

Answer

A. 14 m. Total distance = 10 + 4 = 14 m. Distance adds the lengths of both parts.

P11. Define a scalar quantity.

Answer

A quantity specified by magnitude and unit without an associated spatial direction, such as mass or distance.

P12. Define a vector quantity.

Answer

A quantity with magnitude and direction, such as displacement or velocity.

P13. Distinguish distance from displacement.

Answer

Distance is the total path length, while displacement is the vector change from initial to final position.

P14. What choices define a one-dimensional coordinate system?

Answer

An origin and a positive direction, along with a consistent position scale.

P15. Does changing the origin change the displacement of a fixed trip?

Answer

No. Both endpoint coordinates shift equally, so their difference is unchanged.

P16. An object starts at x = -6 m and moves 15 m in the positive direction. Calculate its final position.

Answer

$x_f = x_i + \Delta x = -6 + 15 = 9$ m.

P17. An object starts at x = -8 m and moves 20 m in the positive direction. Calculate its final position.

Answer

$x_f = x_i + \Delta x = -8 + 20 = 12$ m.

P18. An object moves from x = 9 m to x = -6 m. Calculate its displacement.

Answer

$\Delta x = x_f - x_i = -6 - 9 = -15$ m. The negative sign denotes the negative coordinate direction.

P19. An object moves from x = 12 m to x = -8 m. Calculate its displacement.

Answer

$\Delta x = x_f - x_i = -8 - 12 = -20$ m. The negative sign denotes the negative coordinate direction.

P20. A student walks 15 m east and then 6 m west. Taking east as positive, calculate the displacement.

Answer

$\Delta x = +15 - 6 = +9$ m, east.

P21. A student walks 20 m east and then 8 m west. Taking east as positive, calculate the displacement.

Answer

$\Delta x = +20 - 8 = +12$ m, east.

P22. An observation starts at t = 6 s and ends at t = 21 s. Calculate the time interval.

Answer

$\Delta t = t_f - t_i = 21 - 6 = 15$ s.

P23. An observation starts at t = 8 s and ends at t = 28 s. Calculate the time interval.

Answer

$\Delta t = t_f - t_i = 28 - 8 = 20$ s.

P24. A student walks 15 m east and then 6 m west. Calculate the total distance travelled.

Answer

Total distance = 15 + 6 = 21 m. Distance adds the lengths of both parts.

P25. A student walks 20 m east and then 8 m west. Calculate the total distance travelled.

Answer

Total distance = 20 + 8 = 28 m. Distance adds the lengths of both parts.

P26. A student walks from x = −4 m to x = +6 m and then to x = +1 m. Find distance and displacement.

Answer

Distance = 10 + 5 = 15 m. Displacement = 1 − (−4) = +5 m.

P27. Give an example of nonzero distance with zero displacement.

Answer

Walking 10 m east and then 10 m west back to the start gives distance 20 m but displacement 0.

P28. Can a negative position be combined with a positive displacement? Explain with numbers.

Answer

Yes. Moving from −8 m to −3 m gives $\Delta x = -3 - (-8) = +5$ m while the final position is still negative.

P29. A student adds the magnitudes of two opposite displacements to find the net displacement. Evaluate the method.

Answer

It ignores direction. Choose a positive direction and add signed displacements; adding magnitudes gives distance only for the specified path.

P30. Redraw a trip after moving the coordinate origin 10 m right. Explain what remains unchanged.

Answer

Subtract 10 m from both endpoint coordinates. The physical trip, elapsed time, path length and displacement remain unchanged.